$$x' = s_x \cdot x$$
$$y' = s_y \cdot y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \cancel{\begin{pmatrix} s_x \\ s_y \end{pmatrix}} \cancel{\begin{pmatrix} x \\ y \end{pmatrix}}$$

2x1 2x1

$$= \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Scaling matrix

$$x' = s_x \cdot x$$
$$y' = s_y \cdot y$$

A
$x' = 2 \times 0 = 0$
$y' = 3 \times 0 = 0$

B
$x' = 2 \times 0 = 0$
$y' = 3 \times 3 = 9$

C
$x' = 2 \times 3 = 6$, $y' = 3 \times 3 = 9$

D
$x' = 2 \times 3 = 6$, $y' = 3 \times 0 = 0$

$A(0,0)$, $B(0,3)$, $C(3,3)$, $D(3,0)$

Scale this square $\underbrace{2}_{s_x}$ in x, $\underbrace{3}_{s_y}$ in y

A'
$$\begin{pmatrix} x'_A \\ y'_A \end{pmatrix} = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix} \begin{pmatrix} x_A \\ y_A \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \checkmark$$

B'
$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 9 \end{pmatrix}$$

C'
$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \end{pmatrix}$$

D'
$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$$

$$\begin{array}{cccc} A' & B' & C' & D' \end{array} \qquad\qquad \begin{array}{cccc} A & B & C & D \end{array}$$
$$\begin{pmatrix} x'_A & x'_B & x'_C & x'_D \\ y'_A & y'_B & y'_C & y'_D \end{pmatrix} = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix} \begin{pmatrix} x_A & x_B & x_C & x_D \\ y_A & y_B & y_C & y_D \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}}_{2\times2} \underbrace{\begin{pmatrix} 0 & 0 & 3 & 3 \\ 0 & 3 & 3 & 0 \end{pmatrix}}_{2\times4}$$

$$= \begin{pmatrix} 0 & 0 & 6 & 6 \\ 0 & 9 & 9 & 0 \end{pmatrix}$$